

Interpolation on a Geometric Comb

Lagrange and Newton Formulas, Jackson Calculus, Modal Filters,
Geometric B-Splines, and Fabius–Rvachev Connections

Research report for the ProveIt Fabius Function project

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Abstract

Let $0 < q < 1$ and let

$$\mathcal{C}(A, q) = \{A, Aq, Aq^2, \dots\}$$

be a geometric comb accumulating at the origin. The linked ProveIt monograph already develops the finite Lagrange row that extrapolates polynomial data on $1, q, \dots, q^n$ to 0, its Gaussian-binomial moment cancellations, and its application to scaled Fourier samples of the Rvachev function. This report builds a more systematic interpolation theory around that row and derives a number of extensions that do not merely repeat the existing formulas.

The main results include: explicit scale-covariant barycentric and Newton formulas on Aq^k ; an exact identification of Newton divided differences with Jackson derivatives; a locally uniformly convergent infinite Newton series for analytic functions; a sharp coefficient-majorant error bound summed by the reciprocal q -binomial theorem; a dilation-operator factorization and an $O(n^2)$ triangular algorithm; unique confluent filters annihilating prescribed Puiseux–logarithmic modes; exact total variation and its modular $q \uparrow 1$ asymptotic; an ℓ^1 boundary-layer limit and a regular-variation theorem; a Hermite–Genocchi remainder represented by a positive geometric-knot B-spline; explicit moments and a scaling limit of that B-spline to a positive random variable whose Laplace transform is $1/(-s; q)_\infty$; finite box-spline approximants to geometric-uniform laws and their Thue–Morse collapse at $q = 1/2$; an exact sign formula and probabilistic remainder identity for Fabius values near the flat endpoint; and a base-squaring principle explaining why the Rvachev Fourier problem naturally uses interpolation base $Q = q^2$.

All statements advertised as theorems are proved. Conjectures and open problems are separated explicitly. A commented Python program reproduces the numerical checks and figures and is included with the source and rendered PDF.

Scope and priority. The report was written after inspecting the linked ProveIt monograph and the nearby primary Fabius/Rvachev exposition in the same repository. “New” below means new relative to those inspected documents; it is not a claim that every result is absent from the wider literature. Classical ingredients such as Newton interpolation, Hermite–Genocchi formulas, regular variation, and B-splines are cited where they enter.

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1 Introduction and contribution map

1.1 Why a geometric comb is different

Classical interpolation is often organized around equally spaced nodes, Chebyshev nodes, or a fixed compact array. A geometric comb has a different geometry. Its nodes accumulate inside the domain, the local mesh ratio is constant rather than the mesh difference, and the dilation operator $f(x) \mapsto f(qx)$ diagonalizes the monomials. Consequently the natural algebra is not ordinary finite-difference algebra but Gaussian-binomial and Jackson q -calculus.

The accumulation point makes the problem look ill conditioned: the full Vandermonde determinant tends to zero extremely rapidly. Yet the particular linear functional “evaluate at the accumulation point” has a bounded total variation for every fixed $q < 1$, uniformly in the interpolation order. This apparent paradox is one of the central themes of the report. Coefficient recovery in the monomial basis is badly conditioned, while the scale-adapted functional is stable because its weights develop a universal boundary layer at the smallest nodes.

The same algebra also occurs in the Fabius/Rvachev setting. The Fabius random variable is an infinite sum of geometrically weighted uniforms; the Rvachev Fourier transform is an infinite product of sinc factors; and scaled Fourier samples of an even function convert the physical scale ratio into its square. Thus interpolation, q -series, probability, compactly supported splines, and self-similar smooth functions are not separate topics here. They are different representations of the same dilation structure.

1.2 Baseline from the linked monograph

The source monograph [12] already contains the following core facts, which are treated here as a baseline rather than as new results:

- (i) Newton interpolation on $1, q, \dots, q^n$ with explicit coefficients;
- (ii) the reversed Gaussian row

$$N_{n,k} = (-1)^{n-k} q^{\binom{n-k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q;$$

- (iii) the normalized origin weights

$$\lambda_{n,k}^{(q)} = \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q; q)_k (q; q)_{n-k}};$$

- (iv) exact annihilation of the powers q^{kd} for $1 \leq d \leq n$ and the Gaussian residual for $d \geq n+1$;
- (v) formal and analytic filters obtained from these moments;
- (vi) geometric-uniform random sums, the Fabius function at $q = 1/2$, and a finite Fourier filter for the Rvachev sinc product.

The present report retains this notation but develops the surrounding theory in a form suited to reuse, asymptotics, and future formalization.

1.3 Results developed here

The principal additions are organized as follows.

Topic	Result developed in this report
Comb geometry	Exact Vandermonde determinant for Aq^k , scale covariance, and barycentric denominators.
Newton theory	Jackson-derivative identification, two inverse Gaussian basis transforms, and a convergent infinite Newton expansion for analytic functions.
Analytic error	Exact Taylor tail and the sharper bound $q^{\binom{n+1}{2}} t^{n+1} / (t; q)_{n+1}$.
Operator theory	Spectral polynomial in the dilation operator, recursive factorization, and unique filters for arbitrary simple or confluent modes.
Asymptotics	Exact total variation, $q \uparrow 1$ modular growth, an ℓ^1 boundary layer, and a theorem for regularly varying germs.
Smooth remainder	Hermite–Genocchi expectation, sign theorem, positive geometric-knot B-spline density, and exact Gaussian-binomial moments.
Scaling limit	Rescaled geometric B-splines converge weakly to $Z_q = \sum_{j \geq 0} q^j E_j$ with Laplace transform $1/(-s; q)_\infty$ and an explicit exponential series for its density.
Fabius function	Exact boundary-comb identity, alternating sign, factorial bound, and a Dirichlet-average interpretation of exact dyadic filters.
Rvachev function	Base-squaring principle $Q = c^2$, entire Newton reconstruction from multiscale samples, and moment-flat residuals.

2 Notation and elementary q -identities

Throughout the main body, $0 < q < 1$, $A > 0$, and $n \in \mathbb{N}_0$. Most algebraic formulas remain valid for complex $A \neq 0$ and $0 < |q| < 1$ after the obvious noncollision assumptions.

Definition 2.1 (q -shifted factorial and Gaussian coefficient). For $m \in \mathbb{N}_0$,

$$(a; q)_m := \prod_{j=0}^{m-1} (1 - aq^j), \quad (a; q)_\infty := \prod_{j=0}^{\infty} (1 - aq^j),$$

and

$$\begin{bmatrix} m \\ k \end{bmatrix}_q := \frac{(q; q)_m}{(q; q)_k (q; q)_{m-k}} \quad (0 \leq k \leq m).$$

We also write

$$[m]_q := \frac{1 - q^m}{1 - q}, \quad [m]_q! := \prod_{j=1}^m [j]_q = \frac{(q; q)_m}{(1 - q)^m}.$$

We repeatedly use the finite and reciprocal q -binomial theorems

$$(z; q)_m = \sum_{k=0}^m (-1)^k q^{\binom{k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q z^k, \tag{2.1}$$

$$\frac{1}{(z; q)_{m+1}} = \sum_{r=0}^{\infty} \begin{bmatrix} m+r \\ m \end{bmatrix}_q z^r, \quad |z| < 1. \tag{2.2}$$

These are standard; see DLMF Chapter 17 [6] or Gasper–Rahman [8].

Let $h_r(y_0, \dots, y_m)$ denote the complete homogeneous symmetric polynomial, characterized by

$$\sum_{r \geq 0} h_r(y_0, \dots, y_m) t^r = \prod_{j=0}^m \frac{1}{1 - y_j t}.$$

A direct use of (2.2) gives

$$h_r(A, Aq, \dots, Aq^m) = A^r \begin{bmatrix} m+r \\ m \end{bmatrix}_q. \quad (2.3)$$

This small identity is the source of nearly every Gaussian residual below.

3 Geometry of the comb

Definition 3.1 (Geometric comb). For $A \neq 0$ and $0 < |q| < 1$, define

$$\mathcal{C}(A, q) := \{Aq^k : k \in \mathbb{N}_0\}, \quad \mathcal{C}_n(A, q) := \{A, Aq, \dots, Aq^n\}.$$

A shifted comb $\{aq^{s+k} : k \geq 0\}$ is the same object with $A = aq^s$.

3.1 Scale covariance

If $x = A\tau$, every Lagrange cardinal polynomial on $\mathcal{C}_n(A, q)$ is a function only of (τ, q) . Thus the scale A is not an essential parameter; it simply transports the theory between neighborhoods of the origin.

Proposition 3.2 (Scale covariance). *Let $\ell_{n,k}^{A,q}$ be the cardinal polynomial for the node Aq^k . Then*

$$\ell_{n,k}^{A,q}(A\tau) = \ell_{n,k}^{1,q}(\tau).$$

In particular, the weights for evaluation at 0 are independent of A and of the shift exponent s .

Proof. In the product

$$\ell_{n,k}^{A,q}(A\tau) = \prod_{\substack{0 \leq j \leq n \\ j \neq k}} \frac{A\tau - Aq^j}{Aq^k - Aq^j},$$

each factor A cancels. □

3.2 Vandermonde determinant

The global monomial-basis interpolation problem becomes rapidly singular. The exact determinant quantifies the effect.

Theorem 3.3 (Geometric Vandermonde determinant). *Let $V_n = (x_i^j)_{0 \leq i, j \leq n}$ with $x_i = Aq^i$, and put $N = \binom{n+1}{2}$. Then*

$$\det V_n = (-1)^N A^N q^{\binom{n+1}{3}} \prod_{d=1}^n (1 - q^d)^{n+1-d}. \quad (3.1)$$

Proof. The usual Vandermonde product gives

$$\det V_n = \prod_{0 \leq i < j \leq n} (Aq^j - Aq^i).$$

For each pair, $Aq^j - Aq^i = -Aq^i(1 - q^{j-i})$. There are N pairs. Moreover

$$\sum_{0 \leq i < j \leq n} i = \sum_{i=0}^{n-1} i(n-i) = \binom{n+1}{3},$$

and a fixed difference $d = j - i$ occurs $n + 1 - d$ times. Multiplication yields (3.1). $\square$

Remark 3.4 (Conditioning is functional-dependent). The factor $q^{\binom{n+1}{3}}$ makes the determinant extremely small. It would be a mistake, however, to infer that every linear functional of the interpolant is comparably unstable. The origin functional has total variation bounded by $(-q; q)_\infty / (q; q)_\infty$ for fixed q ; see section 8. The monomial coefficients are ill conditioned, while a scale-adapted extrapolated value need not be.

4 Lagrange interpolation on a finite comb

4.1 Cardinal functions and barycentric weights

Define the node polynomial

$$\omega_{n,A,q}(x) := \prod_{j=0}^n (x - Aq^j) = x^{n+1}(A/x; q)_{n+1},$$

where the last expression is interpreted by polynomial continuation at $x = 0$.

Theorem 4.1 (Explicit denominator and cardinal function). *For $0 \leq k \leq n$,*

$$d_{n,k}^{A,q} := \prod_{\substack{0 \leq j \leq n \\ j \neq k}} (Aq^k - Aq^j) = (-1)^k A^n q^{kn-k(k+1)/2} (q; q)_k (q; q)_{n-k}. \quad (4.1)$$

Consequently

$$\ell_{n,k}^{A,q}(x) = \frac{\omega_{n,A,q}(x)}{(x - Aq^k)d_{n,k}^{A,q}}, \quad (4.2)$$

with the removable value 1 at $x = Aq^k$.

Proof. For $j < k$,

$$Aq^k - Aq^j = -Aq^j(1 - q^{k-j}),$$

whose product is $(-1)^k A^k q^{k(k-1)/2} (q; q)_k$. For $j > k$ the product is $A^{n-k} q^{k(n-k)} (q; q)_{n-k}$. Multiplying gives (4.1); (4.2) is the standard cardinal formula. $\square$

The first-form barycentric weights are $b_{n,k}^{A,q} = 1/d_{n,k}^{A,q}$. For x not equal to a node, the interpolant of data y_k can therefore be computed as

$$\mathcal{I}_n y(x) = \frac{\sum_{k=0}^n \frac{b_{n,k}^{A,q} y_k}{x - Aq^k}}{\sum_{k=0}^n \frac{b_{n,k}^{A,q}}{x - Aq^k}}. \quad (4.3)$$

This avoids solving the Vandermonde system and is the recommended formula away from the accumulation point.

4.2 The origin row

Corollary 4.2 (Scale-free origin weights). *For polynomial data of degree at most n ,*

$$p(0) = \sum_{k=0}^n \lambda_{n,k}^{(q)} p(Aq^k),$$

where

$$\lambda_{n,k}^{(q)} = \ell_{n,k}^{A,q}(0) = \frac{(-1)^{n-k} q^{\binom{n-k+1}{2}}}{(q; q)_k (q; q)_{n-k}}. \quad (4.4)$$

Equivalently,

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} z^k = W_{n,q}(z) := \frac{\prod_{j=1}^n (z - q^j)}{(q; q)_n}. \quad (4.5)$$

Proof. Set $x = 0$ in the product definition of the cardinal function and simplify with (4.1). Formula (4.5) follows from the finite q -binomial theorem or by observing that both sides are degree- n polynomials with the same roots and value 1 at $z = 1$. $\square$

4.3 Residuals at an arbitrary target

The usual interpolation remainder becomes especially explicit on a geometric comb.

Theorem 4.3 (Complete-homogeneous residual). *Let $r \geq 0$. Interpolation of the monomial u^{n+1+r} at $A, Aq, \dots, Aq^n$, evaluated at x , satisfies*

$$\begin{aligned} \sum_{k=0}^n \ell_{n,k}^{A,q}(x) (Aq^k)^{n+1+r} &= x^{n+1+r} - \omega_{n,A,q}(x) h_r(x, A, Aq, \dots, Aq^n) \\ &= x^{n+1+r} - \omega_{n,A,q}(x) \sum_{j=0}^r x^j A^{r-j} \begin{bmatrix} n+r-j \\ n \end{bmatrix}_q. \end{aligned} \quad (4.6)$$

Proof. For a monomial of degree $n+1+r$, the divided difference of order $n+1$ at $x, A, Aq, \dots, Aq^n$ is the complete homogeneous polynomial $h_r(x, A, Aq, \dots, Aq^n)$. The standard remainder identity is therefore the first line. Splitting a complete homogeneous polynomial according to the power of the distinguished variable x and using (2.3) gives the second line. $\square$

At $x = 0$ this gives the exact moment row.

Corollary 4.4 (All origin moments). *For $d \in \mathbb{N}_0$,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} (Aq^k)^d = \begin{cases} 1, & d = 0, \\ 0, & 1 \leq d \leq n, \\ (-1)^n A^d q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \geq n+1. \end{cases} \quad (4.7)$$

4.4 Exact analytic tail and a sharp majorant

For a formal or convergent power series $f(x) = \sum_{m \geq 0} a_m x^m$, define

$$(\mathcal{G}_{n,q} f)(A) := \sum_{k=0}^n \lambda_{n,k}^{(q)} f(Aq^k). \quad (4.8)$$

Then (4.7) gives the exact identity

$$(\mathcal{G}_{n,q}f)(A) - f(0) = (-1)^n q^{\binom{n+1}{2}} \sum_{r=0}^{\infty} \begin{bmatrix} n+r \\ n \end{bmatrix}_q a_{n+1+r} A^{n+1+r}. \quad (4.9)$$

Theorem 4.5 (Reciprocal- q -binomial majorant). *Suppose f is analytic in $|z| < R$ and its Taylor coefficients satisfy $|a_m| \leq MR^{-m}$. Put $t = |A|/R < 1$. Then*

$$|(\mathcal{G}_{n,q}f)(A) - f(0)| \leq M q^{\binom{n+1}{2}} \frac{t^{n+1}}{(t; q)_{n+1}}. \quad (4.10)$$

The right side is coefficientwise sharp: equality in the majorant sum is attained when the relevant coefficients have a common phase and magnitude MR^{-m} .

Proof. Take absolute values in (4.9) and apply (2.2):

$$\sum_{r \geq 0} \begin{bmatrix} n+r \\ n \end{bmatrix}_q t^r = \frac{1}{(t; q)_{n+1}}.$$

□

For fixed $q < 1$ and $t < 1$, $(t; q)_{n+1}$ tends to the positive constant $(t; q)_{\infty}$. Thus the dominant convergence factor is the supergeometric $q^{n(n+1)/2}$, multiplied by the ordinary t^{n+1} .

5 Newton interpolation and Jackson calculus

5.1 Jackson derivative as a divided difference

Define the Jackson derivative

$$(D_q f)(x) := \frac{f(x) - f(qx)}{(1-q)x}, \quad x \neq 0. \quad (5.1)$$

For analytic f , the value at $x = 0$ is supplied by continuity.

Theorem 5.1 (Geometric divided differences). *For every polynomial f and $k \geq 0$,*

$$f[A, Aq, \dots, Aq^k] = \frac{D_q^k f(A)}{[k]_q!}. \quad (5.2)$$

The identity extends to analytic f wherever the divided difference is defined.

Proof. Both sides are linear in f , so it suffices to take $f(x) = x^m$. If $m < k$, both sides vanish. If $m = k + r$, the divided difference is

$$h_r(A, Aq, \dots, Aq^k) = A^r \begin{bmatrix} k+r \\ k \end{bmatrix}_q.$$

On the other hand,

$$D_q^k x^{k+r} \big|_{x=A} = [k+r]_q [k+r-1]_q \cdots [r+1]_q A^r,$$

and division by $[k]_q!$ gives the same Gaussian coefficient. □

5.2 Newton formula and explicit coefficients

Theorem 5.2 (Newton–Jackson interpolation). *The degree- n interpolant of f on $A, Aq, \dots, Aq^n$ is*

$$\mathcal{I}_n f(x) = \sum_{k=0}^n \frac{D_q^k f(A)}{[k]_q!} \prod_{r=0}^{k-1} (x - Aq^r). \quad (5.3)$$

The coefficient may also be recovered directly from samples:

$$\frac{D_q^k f(A)}{[k]_q!} = A^{-k} \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} f(Aq^j). \quad (5.4)$$

Proof. The first formula is ordinary Newton interpolation together with (5.2). For the second, write a divided difference as the sum of values divided by the products of node differences and use (4.1) with $n = k$. $\square$

5.3 Two inverse Gaussian basis transforms

Let

$$\phi_k(x) := \prod_{r=0}^{k-1} (x - Aq^r), \quad \phi_0 = 1.$$

Theorem 5.3 (Monomial–Newton transforms). *For $k, m \in \mathbb{N}_0$,*

$$\phi_k(x) = \sum_{j=0}^k (-1)^{k-j} A^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q x^j, \quad (5.5)$$

$$x^m = \sum_{k=0}^m A^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q \phi_k(x). \quad (5.6)$$

Thus the two triangular coefficient matrices are exact inverses.

Proof. Equation (5.5) is the finite q -binomial theorem applied to $x^k(A/x; q)_k$. For (5.6), use the Newton formula for the monomial x^m and (5.2). The coefficient is $A^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q$. $\square$

If $f(x) = \sum_{m \geq 0} a_m x^m$, comparison with (5.6) gives a useful coefficient transform:

$$c_k := \frac{D_q^k f(A)}{[k]_q!} = \sum_{r=0}^{\infty} a_{k+r} A^r \begin{bmatrix} k+r \\ k \end{bmatrix}_q. \quad (5.7)$$

This is a Gaussian-binomial tail transform of the Taylor coefficients.

5.4 Infinite Newton reconstruction

Because the nodes accumulate at an interior point, analytic data on the comb determine the entire analytic germ. The following theorem makes that uniqueness constructive.

Theorem 5.4 (Locally uniform infinite Newton series). *Let f be analytic in the disk $|x| < R$, let $0 < A < R$, and let $0 < q < 1$. Then*

$$f(x) = \sum_{k=0}^{\infty} \frac{D_q^k f(A)}{[k]_q!} \prod_{r=0}^{k-1} (x - Aq^r) \quad (5.8)$$

locally uniformly for $|x| < R$. At the accumulation point,

$$f(0) = \sum_{k=0}^{\infty} \frac{(-A)^k q^{\binom{k}{2}}}{[k]_q!} D_q^k f(A). \quad (5.9)$$

Proof. Fix radii $0 < r < \rho < R$ with $A < \rho$. Cauchy's estimate gives $|a_m| \leq M_\rho \rho^{-m}$, where $M_\rho = \max_{|z|=\rho} |f(z)|$. From (5.7) and the uniform boundedness of Gaussian coefficients for fixed q ,

$$|c_k| \leq C_{q,A,\rho} M_\rho \rho^{-k}.$$

Choose $\varepsilon > 0$ so that $r + \varepsilon < \rho$. For all sufficiently large j , $Aq^j \leq \varepsilon$. Splitting off finitely many factors gives a constant C such that, uniformly for $|x| \leq r$,

$$\left| \prod_{j=0}^{k-1} (x - Aq^j) \right| \leq C(r + \varepsilon)^k.$$

The Weierstrass test proves local uniform convergence to an analytic function g . At $x = Aq^m$, every term with $k > m$ vanishes, while the first $m + 1$ terms form the Newton interpolant through that node. Hence $g(Aq^m) = f(Aq^m)$ for all m . The nodes accumulate at 0 inside the disk, so the identity theorem gives $g = f$. Setting $x = 0$ yields (5.9). $\square$

Remark 5.5. Absolute and uniform convergence of Jackson q -Taylor/interpolation series has also been studied in the broader q -calculus literature; see Annaby–Mansour [1]. The proof above is tailored to the geometric comb and emphasizes the identity theorem at its accumulation point.

6 Dilation-operator calculus and the Gaussian filter

6.1 Spectral polynomial

Let T_q denote the dilation operator

$$(T_q f)(x) := f(qx).$$

Monomials are its eigenfunctions: $T_q x^m = q^m x^m$. The origin filter is therefore a polynomial functional calculus for T_q .

Theorem 6.1 (Operator factorization). *For every function for which the samples exist,*

$$\mathcal{G}_{n,q} = W_{n,q}(T_q) = \frac{1}{(q; q)_n} \prod_{j=1}^n (T_q - q^j I). \quad (6.1)$$

Equivalently,

$$\mathcal{G}_{0,q} = I, \quad \mathcal{G}_{n,q} = \frac{T_q - q^n I}{1 - q^n} \mathcal{G}_{n-1,q}. \quad (6.2)$$

Proof. Insert T_q into the row polynomial (4.5). The recursion is obtained by separating the final factor. $\square$

The factorization gives a triangular algorithm that never forms the full row. If $y_k = f(Aq^k)$, replace adjacent entries recursively by

$$y_k \leftarrow \frac{y_{k+1} - q^j y_k}{1 - q^j}, \quad j = 1, 2, \dots, n. \quad (6.3)$$

After level j , the array is shorter by one; after level n , its single entry is $(\mathcal{G}_{n,q}f)(A)$. The arithmetic cost is $n(n+1)/2$ updates and $O(n)$ storage.

Corollary 6.2 (Weight recurrence). *With the convention $\lambda_{m,k} = 0$ outside $0 \leq k \leq m$,*

$$\lambda_{n,k}^{(q)} = \frac{\lambda_{n-1,k-1}^{(q)} - q^n \lambda_{n-1,k}^{(q)}}{1 - q^n}. \quad (6.4)$$

6.2 Exact action on powers and logarithmic powers

The spectral point of x^α is q^α , even when α is not an integer.

Theorem 6.3 (Power multiplier). *For $A > 0$ and $\alpha \in \mathbb{C}$,*

$$(\mathcal{G}_{n,q}x^\alpha)(A) = A^\alpha W_{n,q}(q^\alpha) = A^\alpha q^{\alpha n} \frac{(q^{1-\alpha}; q)_n}{(q; q)_n}. \quad (6.5)$$

For every integer $m \in \{1, \dots, n\}$ this is zero.

Proof. Since $(Aq^k)^\alpha = A^\alpha (q^k)^\alpha$, the weighted sum is $A^\alpha W_{n,q}(q^\alpha)$. Factoring q^α from each factor of $W_{n,q}$ gives the second expression. $\square$

Differentiating with respect to the exponent yields all logarithmic companions.

Corollary 6.4 (Logarithmic modes). *For $s \in \mathbb{N}_0$,*

$$(\mathcal{G}_{n,q}[x^\alpha (\log x)^s])(A) = \partial_\alpha^s \left[A^\alpha q^{\alpha n} \frac{(q^{1-\alpha}; q)_n}{(q; q)_n} \right]. \quad (6.6)$$

A simple zero of $W_{n,q}$ at q^m annihilates x^m but generally not $x^m \log x$. This observation motivates confluent spectral filters.

6.3 Uniqueness

Proposition 6.5 (Minimum-degree uniqueness). *Among all filters $\mathcal{R}f(x) = \sum_{k=0}^n w_k f(q^k x)$ that preserve constants and annihilate $x, x^2, \dots, x^n$, the Gaussian row is unique.*

Proof. Let $W(z) = \sum_{k=0}^n w_k z^k$. Constant preservation says $W(1) = 1$ and power annihilation says $W(q^j) = 0$ for $1 \leq j \leq n$. The unique degree-at-most- n polynomial with these values is $\prod_{j=1}^n (z - q^j) / (1 - q^j) = W_{n,q}(z)$. $\square$

7 Confluent modal extrapolation

Ordinary polynomial cancellation assumes an integer-power Taylor expansion. Endpoint asymptotics in differential equations, Mellin analysis, and inverse special functions often contain fractional powers and logarithms. The dilation-operator viewpoint gives the correct generalization immediately.

Definition 7.1 (Mode polynomial). Let $\alpha_1, \dots, \alpha_J \in \mathbb{C}$ have pairwise distinct spectral points q^{α_j} , none equal to 1, and let $m_1, \dots, m_J \in \mathbb{N}$, and put $N = \sum_j m_j$. Define

$$W_{\alpha, \mathbf{m}}(z) := \prod_{j=1}^J \left(\frac{z - q^{\alpha_j}}{1 - q^{\alpha_j}} \right)^{m_j} = \sum_{k=0}^N w_k z^k, \quad (7.1)$$

and

$$(\mathcal{R}_{\alpha, \mathbf{m}} f)(x) := \sum_{k=0}^N w_k f(q^k x).$$

Theorem 7.2 (Confluent annihilation). *The filter (7.1) preserves constants and, for every j and $0 \leq s < m_j$, annihilates*

$$x^{\alpha_j} (\log x)^s.$$

It is the unique filter of degree at most N with these properties.

Proof. For a general polynomial $W(z) = \sum_k w_k z^k$,

$$\begin{aligned} \sum_k w_k (q^k x)^\alpha (\log(q^k x))^s &= x^\alpha \sum_k w_k q^{k\alpha} (\log x + k \log q)^s \\ &= x^\alpha (\log x + \partial_\alpha)^s W(q^\alpha). \end{aligned}$$

A zero of multiplicity m_j at q^{α_j} makes the right side vanish for $s < m_j$. The normalization $W(1) = 1$ preserves constants. Conversely, the annihilation equations imply zeros with the stated multiplicities; degree and normalization then force (7.1). $\square$

Corollary 7.3 (Exact general multiplier). *For any $\alpha \in \mathbb{C}$ and $s \in \mathbb{N}_0$,*

$$\mathcal{R}_{\alpha, \mathbf{m}}[x^\alpha (\log x)^s] = x^\alpha (\log x + \partial_\alpha)^s W_{\alpha, \mathbf{m}}(q^\alpha). \quad (7.2)$$

Theorem 7.4 (Puiseux–log acceleration). *Suppose, as $x \downarrow 0$,*

$$f(x) = a_0 + \sum_{j=1}^J \sum_{s=0}^{m_j-1} a_{j,s} x^{\alpha_j} (\log x)^s + O\left(x^\beta (1 + |\log x|^r)\right), \quad (7.3)$$

where $\Re \alpha_j > 0$ and $\beta > 0$. Then

$$(\mathcal{R}_{\alpha, \mathbf{m}} f)(x) = a_0 + O\left(x^\beta (1 + |\log x|^r)\right). \quad (7.4)$$

Proof. The finite sum of displayed modes is annihilated exactly. For each fixed $k \leq N$, replacing x by $q^k x$ preserves the stated remainder order, with a constant depending on q, k, β, r . Summing finitely many weighted remainders proves the assertion. $\square$

Remark 7.5 (Relation to Richardson extrapolation). This is geometric Richardson extrapolation in its natural spectral form. The usual integer-power Gaussian filter corresponds to $\alpha_j = j$, $m_j = 1$. Repeated exponents handle logarithmic resonance without introducing derivative data at the nodes.

Problem 7.6 (Stability-optimal modal design). *For a prescribed list of modes, (7.1) is algebraically forced at minimum degree. If extra degrees of freedom are allowed, determine the polynomial W with $W(1) = 1$ and the required confluent zeros that minimizes $\sum_k |w_k|$, or a weighted noise norm adapted to the expected precision of $f(q^k x)$. This is a finite convex optimization problem whose asymptotic structure appears unexplored in the Fabius/Rvachev setting.*

8 Stability, boundary layers, and regular variation

8.1 Exact total variation

Because the signs of (4.4) alternate, the total variation can be read from the row polynomial at -1 .

Theorem 8.1 (Exact noise amplification). *For $0 < q < 1$,*

$$K_n(q) := \sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-q; q)_n}{(q; q)_n} = \prod_{j=1}^n \frac{1+q^j}{1-q^j}. \quad (8.1)$$

Consequently

$$K_n(q) \uparrow K_\infty(q) := \frac{(-q; q)_\infty}{(q; q)_\infty} < \infty. \quad (8.2)$$

If each sample has absolute error at most δ , the extrapolated absolute error is at most $K_n(q)\delta$.

Proof. Since $\text{sgn } \lambda_{n,k} = (-1)^{n-k}$,

$$\sum_k |\lambda_{n,k}| = (-1)^n \sum_k \lambda_{n,k} (-1)^k = (-1)^n W_{n,q}(-1).$$

Substitute (4.5). Monotone convergence of the positive product gives (8.2). $\square$

For reference,

$$K_\infty(1/4) \approx 1.9692603537, \quad K_\infty(1/2) \approx 8.2559879358, \quad K_\infty(3/4) \approx 803.0132718.$$

Thus the Fourier base $Q = 1/4$ associated with dyadic Rvachev scaling is much better conditioned than a direct dyadic $q = 1/2$ extrapolation.

8.2 The singular regime $q \uparrow 1$

Two limits must be distinguished. For fixed n ,

$$K_n(q) \sim \frac{2^n}{n!(1-q)^n} \quad (q \uparrow 1). \quad (8.3)$$

If the order tends to infinity first, the growth is nonperturbative.

Theorem 8.2 (Modular asymptotic). *Write $q = e^{-t}$ with $t \downarrow 0$. Then*

$$K_\infty(e^{-t}) \sim \frac{\sqrt{t}}{2\sqrt{\pi}} \exp\left(\frac{\pi^2}{4t}\right). \quad (8.4)$$

Proof. Use $(-q; q)_\infty = (q^2; q^2)_\infty / (q; q)_\infty$, so

$$K_\infty(q) = \frac{(q^2; q^2)_\infty}{(q; q)_\infty^2}.$$

The standard Dedekind-eta asymptotic

$$(e^{-t}; e^{-t})_\infty \sim \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right)$$

applied at t and $2t$ gives (8.4); the elementary exponential corrections cancel. $\square$

The plot in figure 1 shows the practical consequence: ratios close to one are unsuitable for high-order accumulation-point extrapolation unless precision grows very rapidly.

8.3 The reversed-row boundary layer

Put $r = n - k$. For fixed r ,

$$\lambda_{n, n-r}^{(q)} \longrightarrow \beta_r^{(q)} := \frac{(-1)^r q^{\binom{r+1}{2}}}{(q; q)_\infty (q; q)_r}. \quad (8.5)$$

The limiting signed generating function is

$$\sum_{r=0}^{\infty} \beta_r^{(q)} z^r = \frac{(qz; q)_\infty}{(q; q)_\infty}. \quad (8.6)$$

In particular, its mass is 1 and its total variation is $K_\infty(q)$.

Theorem 8.3 (ℓ^1 boundary-layer convergence). *Extend the reversed row by zero for $r > n$. Then*

$$\sum_{r=0}^{\infty} \left| \lambda_{n, n-r}^{(q)} - \beta_r^{(q)} \right| \longrightarrow 0. \quad (8.7)$$

Proof. Pointwise convergence follows from $(q; q)_{n-r} \rightarrow (q; q)_\infty$. After multiplication by $(-1)^r$, all terms are nonnegative. Their total masses are $K_n(q)$ and $K_\infty(q)$, respectively, and the former tends to the latter. Scheffe's lemma for counting measure gives convergence in ℓ^1 . $\square$

This theorem explains the stability mechanism. For large n , the effective weights sit a bounded number of steps behind the smallest node Aq^n ; the large-node coefficients are superexponentially small.

8.4 Convergence for merely continuous data

Analyticity is not needed for consistency.

Theorem 8.4 (Continuous accumulation-point convergence). *If f is continuous on $[0, A]$, then*

$$(\mathcal{G}_{n,q}f)(A) \longrightarrow f(0). \quad (8.8)$$

More quantitatively, if $\omega_f(\delta) = \sup_{0 \leq x \leq \delta} |f(x) - f(0)|$, then for $1 \leq J \leq n+1$,

$$\begin{aligned} |(\mathcal{G}_{n,q}f)(A) - f(0)| &\leq K_\infty(q) \omega_f(Aq^J) \\ &\quad + \frac{2\|f\|_\infty J}{(q; q)_\infty^2} q^{\binom{n-J+2}{2}}. \end{aligned} \quad (8.9)$$

Proof. Split the weighted sum into $k \geq J$ and $k < J$. On the first part, every node is at most Aq^J , so the total is bounded by $K_n \omega_f(Aq^J)$. On the second part,

$$|\lambda_{n,k}^{(q)}| \leq \frac{q^{\binom{n-k+1}{2}}}{(q; q)_\infty^2} \leq \frac{q^{\binom{n-J+2}{2}}}{(q; q)_\infty^2}.$$

There are at most J such terms, and $|f(Aq^k) - f(0)| \leq 2\|f\|_\infty$. First fix J and let $n \rightarrow \infty$, then let $J \rightarrow \infty$. $\square$

8.5 Regularly varying endpoint germs

A function f is regularly varying at $0+$ with index $\alpha \in \mathbb{R}$ if $f(tx)/f(t) \rightarrow x^\alpha$ as $t \downarrow 0$ for every $x > 0$, with f eventually of one sign. Standard background is in [3].

Theorem 8.5 (Regular-variation transfer). *Let f be regularly varying at $0+$ with index α . Then, for fixed $A > 0$,*

$$\frac{(\mathcal{G}_{n,q}f)(A)}{f(Aq^n)} \longrightarrow C_q(\alpha) := \frac{(q^{1-\alpha}; q)_\infty}{(q; q)_\infty}. \quad (8.10)$$

For a positive integer $\alpha = m$, the limit is zero because the numerator product contains the factor $1 - q^0$.

Proof. Reverse the row:

$$\frac{(\mathcal{G}_{n,q}f)(A)}{f(Aq^n)} = \sum_{r=0}^n \lambda_{n,n-r}^{(q)} \frac{f(Aq^{n-r})}{f(Aq^n)}.$$

For fixed r , regular variation gives a limit $q^{-\alpha r}$, while (8.5) gives the coefficient limit. Potter bounds dominate the ratio by $Cq^{-(\alpha+\varepsilon)r}$ away from a finite initial segment; the factor $q^{r(r+1)/2}$ in β_r makes the resulting series summable for every α . The finitely many terms corresponding to nodes bounded away from zero have superexponentially small coefficients. Dominated convergence and (8.6) now give

$$\sum_{r \geq 0} \beta_r^{(q)} q^{-\alpha r} = \frac{(q^{1-\alpha}; q)_\infty}{(q; q)_\infty}.$$

$\square$

For pure powers, (8.10) is already visible in the exact formula (6.5). The theorem says that slowly varying factors do not change the limiting multiplier.

9 Smooth remainders and geometric-knot B-splines

9.1 Hermite–Genocchi representation

The exact Taylor tail is best for analytic data. For finite smoothness, a divided-difference representation reveals positivity and sign.

Theorem 9.1 (Geometric Hermite–Genocchi remainder). *Let $f \in C^{n+1}[0, A]$ and let $(T_*, T_0, \dots, T_n)$ be uniformly distributed on the $(n+1)$ -simplex, that is, Dirichlet with all parameters equal to 1. Put*

$$Y_{n,A,q} := A \sum_{k=0}^n q^k T_k. \quad (9.1)$$

Then

$$(\mathcal{G}_{n,q}f)(A) - f(0) = \frac{(-1)^n A^{n+1} q^{\binom{n+1}{2}}}{(n+1)!} \mathbb{E}[f^{(n+1)}(Y_{n,A,q})]. \quad (9.2)$$

Proof. Let $I_n f$ be the interpolating polynomial through $A, Aq, \dots, Aq^n$. The ordinary interpolation remainder at 0 gives

$$I_n f(0) - f(0) = (-1)^n A^{n+1} q^{\binom{n+1}{2}} f[0, A, Aq, \dots, Aq^n].$$

The Hermite–Genocchi formula for a divided difference of order $n+1$ is

$$f[0, A, Aq, \dots, Aq^n] = \frac{1}{(n+1)!} \mathbb{E} f^{(n+1)} \left(A \sum_{k=0}^n q^k T_k \right).$$

Since $I_n f(0) = (\mathcal{G}_{n,q}f)(A)$, the result follows. $\square$

Corollary 9.2 (Sharp derivative bound and sign). *Under the hypotheses of the theorem,*

$$|(\mathcal{G}_{n,q}f)(A) - f(0)| \leq \frac{A^{n+1} q^{\binom{n+1}{2}}}{(n+1)!} \|f^{(n+1)}\|_{L^\infty(0,A)}. \quad (9.3)$$

If $f^{(n+1)}$ has a fixed nonzero sign on $(0, A)$, then the remainder has the sign of $(-1)^n f^{(n+1)}$.

9.2 Peano kernel and a positive spline density

For $n \geq 1$, define the Peano kernel

$$K_{n,A,q}(t) := \frac{1}{n!} \sum_{k=0}^n \lambda_{n,k}^{(q)} (Aq^k - t)_+^n, \quad 0 \leq t \leq A. \quad (9.4)$$

Then

$$(\mathcal{G}_{n,q}f)(A) - f(0) = \int_0^A f^{(n+1)}(t) K_{n,A,q}(t) dt.$$

Normalize it by

$$B_{n,A,q}(t) := \frac{(-1)^n (n+1)!}{A^{n+1} q^{\binom{n+1}{2}}} K_{n,A,q}(t). \quad (9.5)$$

Theorem 9.3 (Geometric-knot B-spline). *The function $B_{n,A,q}$ is the probability density of $Y_{n,A,q}$ in (9.1). In particular,*

- (i) $B_{n,A,q} \geq 0$ and is supported on $[0, A]$;
- (ii) $\int_0^A B_{n,A,q}(t) dt = 1$;
- (iii) *it is the classical univariate B-spline associated with the knots $0, Aq^n, \dots, Aq, A$, up to the displayed normalization;*
- (iv) *it is log-concave and hence unimodal on the interior of its support.*

Proof. Comparing the Peano representation with (9.2) for arbitrary continuous $f^{(n+1)}$ identifies the normalized kernel with the law of $Y_{n,A,q}$. This is also the standard Curry–Schoenberg interpretation of a B-spline as the density of a Dirichlet average of its knots; see [5]. Positivity, support, and normalization follow immediately. The uniform density on a simplex is log-concave, and linear images preserve log-concavity by the Prekopa theorem, proving the last assertion. $\square$

Remark 9.4 (Not the same as a “quantum B-spline”). The density above is a classical B-spline whose knots happen to be geometric. It should be distinguished from the quantum B-splines of Simeonov–Goldman [14], where continuity conditions are formulated using q -derivatives. The two theories meet through divided differences and Jackson calculus, but their definitions are different. The geometric-knot density is the one canonically forced by the interpolation remainder in this report.

9.3 Exact moments

Theorem 9.5 (Gaussian-binomial B-spline moments). *For every $r \in \mathbb{N}_0$,*

$$\mathbb{E}Y_{n,A,q}^r = \frac{r!(n+1)!}{(n+r+1)!} A^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q. \quad (9.6)$$

Equivalently,

$$\int_0^A t^r B_{n,A,q}(t) dt = \frac{r!(n+1)!}{(n+r+1)!} A^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q. \quad (9.7)$$

Proof. For a Dirichlet($1, \dots, 1$) vector with $n+2$ components,

$$\mathbb{E} \left(\sum_i x_i T_i \right)^r = \frac{r!(n+1)!}{(n+r+1)!} h_r(x_0, \dots, x_{n+1}).$$

Take the knots $0, A, Aq, \dots, Aq^n$ and apply (2.3). □

Thus the Gaussian coefficient is not only an interpolation residual: it is the moment sequence of the positive Peano-kernel distribution after a simple beta normalization.

9.4 A q -Pochhammer scaling limit

The geometric B-spline has a nontrivial boundary-scale limit. Let $E_*, E_0, E_1, \dots$ be independent exponential random variables of mean 1.

Theorem 9.6 (Scaling limit of the geometric B-spline). *For fixed $0 < q < 1$,*

$$\frac{n+2}{A} Y_{n,A,q} \xrightarrow{d} Z_q := \sum_{k=0}^{\infty} q^k E_k. \quad (9.8)$$

The limit has moments and Laplace transform

$$\mathbb{E}Z_q^r = \frac{r!}{(q; q)_r}, \quad (9.9)$$

$$\mathbb{E}e^{-sZ_q} = \frac{1}{(-s; q)_{\infty}}, \quad s \geq 0. \quad (9.10)$$

For $y > 0$ its density admits the absolutely convergent expansion

$$p_q(y) = \frac{1}{(q; q)_{\infty}} \sum_{k=0}^{\infty} \frac{(-1)^k q^{\binom{k}{2}}}{(q; q)_k} e^{-q^{-k}y}. \quad (9.11)$$

Proof. Use the exponential representation of a uniform Dirichlet vector:

$$(T_*, T_0, \dots, T_n) \stackrel{d}{=} \frac{(E_*, E_0, \dots, E_n)}{E_* + E_0 + \dots + E_n}.$$

The denominator divided by $n + 2$ converges almost surely to 1, while $\sum_{k=0}^n q^k E_k$ converges almost surely and in every L^r to Z_q . Slutsky's theorem proves (9.8). Independence gives

$$\mathbb{E}e^{-sZ_q} = \prod_{k \geq 0} (1 + sq^k)^{-1} = (-s; q)_\infty^{-1}.$$

Expanding $1/(t; q)_\infty = \sum_{r \geq 0} t^r / (q; q)_r$ around $t = 0$ and comparing moment-generating coefficients gives (9.9).

For the density, decompose the Laplace transform into simple fractions at its poles $s = -q^{-k}$:

$$\frac{1}{(-s; q)_\infty} = \sum_{k \geq 0} \frac{(-1)^k q^{\binom{k+1}{2}}}{(q; q)_k (q; q)_\infty} \frac{1}{1 + sq^k}.$$

Termwise inversion gives (9.11); for each $y > 0$ the Gaussian factor in k and the term $e^{-q^{-k}y}$ ensure absolute convergence. $\square$

Remark 9.7. The limit satisfies the perpetuity equation

$$Z_q \stackrel{d}{=} E_0 + qZ'_q,$$

where Z'_q is an independent copy. The q -Pochhammer product is therefore both an interpolation object and the Laplace transform of the boundary-scale limit of the associated B-splines.

10 Geometric-uniform random sums and finite splines

10.1 The generic law

Let $0 < b < 1$ and let $U_0, U_1, \dots$ be independent uniforms on $[0, 1]$. Define

$$X_b := (1 - b) \sum_{j=0}^{\infty} b^j U_j. \tag{10.1}$$

The weights sum to 1, so $X_b \in [0, 1]$. Splitting off the first summand gives the distributional self-similarity

$$X_b \stackrel{d}{=} (1 - b)U_0 + bX'_b, \tag{10.2}$$

where X'_b is an independent copy. If F_b is the distribution function, then

$$F_b(x) = \int_0^1 F_b\left(\frac{x - (1 - b)u}{b}\right) du, \tag{10.3}$$

with the conventions $F_b = 0$ to the left of 0 and $F_b = 1$ to the right of 1. At $b = 1/2$, this is the Fabius distribution.

10.2 Finite generalized Irwin–Hall splines

Let

$$X_{b,N} := (1-b) \sum_{j=0}^{N-1} b^j U_j, \quad a_j = (1-b)b^j.$$

This is a convolution of uniforms on intervals of geometrically decreasing width.

Theorem 10.1 (Finite box-spline formula). *For $N \geq 1$, the density and distribution function of $X_{b,N}$ are*

$$f_{b,N}(x) = \frac{1}{(N-1)! \prod_{j=0}^{N-1} a_j} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=0}^{N-1} \varepsilon_j a_j \right)_+^{N-1}, \quad (10.4)$$

$$F_{b,N}(x) = \frac{1}{N! \prod_{j=0}^{N-1} a_j} \sum_{\varepsilon \in \{0,1\}^N} (-1)^{|\varepsilon|} \left(x - \sum_{j=0}^{N-1} \varepsilon_j a_j \right)_+^N, \quad (10.5)$$

where

$$\prod_{j=0}^{N-1} a_j = (1-b)^N b^{N(N-1)/2}. \quad (10.6)$$

Proof. The density of a uniform on $[0, a]$ is $a^{-1}(H(x) - H(x-a))$, where H is the Heaviside function. Convolving N such factors and expanding the product of translation differences gives the alternating subset sum. The N -fold convolution of H is $x_+^{N-1}/(N-1)!$. Integration gives the CDF, and (10.6) is immediate. $\square$

For a general b , the knots are subset sums of a geometric progression. At the dyadic value they collapse to an arithmetic comb, and the coefficient sign is the Thue–Morse sign.

Corollary 10.2 (Dyadic Thue–Morse collapse). *For $b = 1/2$,*

$$F_{1/2,N}(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{r=0}^{2^N-1} (-1)^{s_2(r)} \left(x - \frac{r}{2^N} \right)_+^N, \quad (10.7)$$

where $s_2(r)$ is the binary digit sum.

Proof. Here $a_j = 2^{-(j+1)}$. Every subset sum is uniquely $r/2^N$, and the subset cardinality has the parity of $s_2(r)$. The product of widths is $2^{-N(N+1)/2}$. $\square$

This identity explains why a geometric collection of widths produces an arithmetic dyadic knot comb: binary representation converts subset selection into position on the uniform grid, while the inclusion–exclusion sign becomes the Thue–Morse sequence.

10.3 Convergence to the infinite law

The omitted tail is bounded deterministically:

$$0 \leq X_b - X_{b,N} \leq b^N.$$

Hence

$$F_b(x) \leq F_{b,N}(x) \leq F_b(x + b^N). \quad (10.8)$$

Because F_b is continuous, $F_{b,N} \rightarrow F_b$ uniformly. Thus the Fabius function is a uniform limit of explicitly known compactly supported splines whose coefficients are a Thue–Morse finite difference.

10.4 Centered law, Bernoulli cumulants, and sinc products

Put

$$Z_b := 2X_b - 1 = 2(1-b) \sum_{j=0}^{\infty} b^j (U_j - 1/2). \quad (10.9)$$

Then Z_b is symmetric and supported on $[-1, 1]$. Under the Fourier convention

$$\widehat{u}_b(z) = \mathbb{E} e^{-2\pi i z Z_b},$$

its characteristic function is

$$\widehat{u}_b(z) = \prod_{j=0}^{\infty} \text{sinc}(2\pi(1-b)b^j z), \quad \text{sinc } w := \frac{\sin w}{w}. \quad (10.10)$$

At $b = 1/2$, this is

$$\widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}(\pi z / 2^j), \quad (10.11)$$

which is the standard Rvachev product.

The centered uniform on $[-1/2, 1/2]$ has cumulants $\kappa_m = B_m/m$ for $m \geq 2$, where B_m are Bernoulli numbers and odd cumulants vanish. Independence and scaling give the following explicit q -denominators.

Proposition 10.3 (Cumulants of the geometric-uniform law). *For $m \geq 1$,*

$$\kappa_{2m}(Z_b) = \frac{[2(1-b)]^{2m} B_{2m}}{2m(1-b^{2m})}, \quad \kappa_{2m+1}(Z_b) = 0. \quad (10.12)$$

Consequently every even moment is an explicit complete Bell polynomial in the numbers (10.12).

Proof. Cumulants add under independent sums and scale homogeneously. Therefore

$$\kappa_{2m}(Z_b) = [2(1-b)]^{2m} \frac{B_{2m}}{2m} \sum_{j \geq 0} b^{2mj},$$

and the geometric sum gives (10.12). □

The factors $1 - b^{2m}$ are the simplest q -analog denominators. In Fourier space the same even degree $2m$ is responsible for the squared interpolation base developed in [section 12](#).

11 Fabius and Rvachev functions at the flat boundary

11.1 Definitions and endpoint differential scaling

Let F be the Fabius distribution function on $[0, 1]$. It is the CDF of $X_{1/2}$ and satisfies

$$F(0) = 0, \quad F(1) = 1, \quad F(1-x) = 1 - F(x), \quad (11.1)$$

and, for $0 \leq x \leq 1/2$,

$$F'(x) = 2F(2x). \quad (11.2)$$

The compactly supported Rvachev function is related by

$$\text{up}(x) = F(1 - |x|), \quad |x| \leq 1, \quad (11.3)$$

with $\text{up}(x) = 0$ for $|x| \geq 1$. These normalizations agree with the primary Fabius/Rvachev literature [7, 13, 2] and with the inspected ProveIt exposition [11].

Repeated differentiation of (11.2) gives the exact endpoint scaling.

Lemma 11.1 (Endpoint derivatives). *For $m \geq 1$ and $0 \leq x \leq 2^{-m}$,*

$$F^{(m)}(x) = 2^{m(m+1)/2} F(2^m x). \quad (11.4)$$

Proof. The case $m = 1$ is (11.2). If the result holds for m , then on $x \leq 2^{-(m+1)}$,

$$F^{(m+1)}(x) = 2^{m(m+1)/2} 2^m F'(2^m x) = 2^{m(m+1)/2+m+1} F(2^{m+1} x),$$

which has exponent $(m+1)(m+2)/2$. $\square$

11.2 An exact boundary-comb identity

The smooth Peano remainder becomes self-similar when $f = F$.

Theorem 11.2 (Fabius boundary filter). *Let $n \geq 0$, $0 < A \leq 2^{-(n+1)}$, and let $(T_*, T_0, \dots, T_n)$ be uniform Dirichlet. Put*

$$S_n := \sum_{k=0}^n 2^{-k} T_k \in [0, 1]. \quad (11.5)$$

Then

$$\sum_{k=0}^n \lambda_{n,k}^{(1/2)} F(A2^{-k}) = \frac{(-1)^n (2A)^{n+1}}{(n+1)!} \mathbb{E} F(2^{n+1} A S_n). \quad (11.6)$$

In particular, the left side has strict sign $(-1)^n$ and obeys

$$0 < (-1)^n \sum_{k=0}^n \lambda_{n,k}^{(1/2)} F(A2^{-k}) \leq \frac{(2A)^{n+1}}{(n+1)!}. \quad (11.7)$$

Proof. Apply (9.2) with $q = 1/2$ and $f = F$. Since $F(0) = 0$,

$$\sum_k \lambda_{n,k}^{(1/2)} F(A2^{-k}) = \frac{(-1)^n A^{n+1} 2^{-n(n+1)/2}}{(n+1)!} \mathbb{E} F^{(n+1)}(A S_n).$$

The hypothesis on A allows (11.4) with $m = n+1$ throughout the support of $A S_n$. The powers of 2 reduce to 2^{n+1} , giving (11.6). The random argument lies in $[0, 1]$, so $0 \leq F \leq 1$. It is positive almost surely because $S_n > 0$ almost surely, which gives strict sign and the bound. $\square$

Corollary 11.3 (Exact dyadic form). *If $M \geq n+1$, then*

$$\sum_{k=0}^n \lambda_{n,k}^{(1/2)} F(2^{-(M+k)}) = \frac{(-1)^n 2^{-(M-1)(n+1)}}{(n+1)!} \mathbb{E} F(2^{n+1-M} S_n). \quad (11.8)$$

For $M = n+1$, the ratio of the absolute value to the upper bound is exactly $\mathbb{E} F(S_n)$.

Through (11.3), the same result is an identity on the left boundary of up:

$$\sum_{k=0}^n \lambda_{n,k}^{(1/2)} \text{up}(-1 + A2^{-k}) = \frac{(-1)^n (2A)^{n+1}}{(n+1)!} \mathbb{E} F(2^{n+1} A S_n). \quad (11.9)$$

Thus a signed combination of values of a nowhere-analytic C^∞ function is represented by a positive expectation and has a completely determined sign.

11.3 The boundary B-spline and its q -exponential scale

The variable S_n is $Y_{n,1,1/2}$ from [section 9](#). Therefore

$$(n+2)S_n \xrightarrow{d} Z_{1/2} = \sum_{k \geq 0} 2^{-k} E_k, \quad \mathbb{E} e^{-sZ_{1/2}} = \frac{1}{(-s; 1/2)_\infty}. \quad (11.10)$$

This identifies the random scale sampled by the normalized Fabius remainder. The typical size of S_n is $2/(n+2)$, but its limiting rescaled distribution is not degenerate; it is the q -Pochhammer perpetuity in [\(9.10\)](#).

11.4 Exact rational computation

All dyadic values of F are rational. A convenient triangular recurrence, also used in the linked monograph, is

$$V_0 = 1, \quad V_m := F(2^{-m}) = \frac{2^{-\binom{m}{2}}}{2^m - 1} \sum_{k=0}^{m-1} \frac{2^{\binom{k}{2}}}{(m-k+1)!} V_k. \quad (11.11)$$

It begins

$$V_0 = 1, \quad V_1 = \frac{1}{2}, \quad V_2 = \frac{5}{72}, \quad V_3 = \frac{1}{288}, \quad V_4 = \frac{143}{2073600}.$$

Because both [\(11.11\)](#) and the dyadic weights $\lambda_{n,k}^{(1/2)}$ are rational, [\(11.8\)](#) can be checked with exact integer arithmetic. The companion Python program does so before switching to multiprecision arithmetic for larger plotted orders.

12 Fourier-scale interpolation for the Rvachev function

12.1 The base-squaring principle

Let Φ be an even entire function normalized by $\Phi(0) = 1$, and write

$$\Phi(z) = \sum_{m=0}^{\infty} c_m z^{2m}. \quad (12.1)$$

Fix a scale ratio $0 < c < 1$ and set

$$Q := c^2, \quad H_z(t) := \Phi(z\sqrt{t}) = \sum_{m=0}^{\infty} c_m z^{2m} t^m. \quad (12.2)$$

Evenness makes H_z entire in t , independent of the choice of square root. Moreover

$$H_z(Q^k) = \Phi(c^k z). \quad (12.3)$$

Therefore geometric interpolation of Fourier scales $c^k z$ is interpolation on the comb $1, Q, Q^2, \dots$, not on base c itself. In the dyadic Rvachev problem, $c = 1/2$ and the interpolation base is $Q = 1/4$.

12.2 Exact moment-flat filter

Let $\lambda_{n,k}^{(Q)}$ denote the origin weights with base Q .

Theorem 12.1 (Even-entire scale filter). *For H_z as in (12.2),*

$$\sum_{k=0}^n \lambda_{n,k}^{(Q)} \Phi(c^k z) = 1 + (-1)^n Q^{\binom{n+1}{2}} \sum_{r=0}^{\infty} \begin{bmatrix} n+r \\ n \end{bmatrix}_Q c_{n+1+r} z^{2n+2+2r}. \quad (12.4)$$

In particular, the residual has a zero of order at least $2n+2$ at $z=0$.

Proof. Apply (4.9) to the entire function $H_z(t)$ at $A=1$ and base Q . $\square$

For a characteristic function under the convention $\Phi(z) = \mathbb{E}e^{-2\pi izX}$ with a symmetric random variable X ,

$$c_m = (-1)^m \frac{(2\pi)^{2m} \mu_{2m}}{(2m)!}, \quad \mu_{2m} = \mathbb{E}X^{2m}. \quad (12.5)$$

Hence the first nonzero term of the filtered residual is always

$$-Q^{\binom{n+1}{2}} \frac{(2\pi z)^{2n+2} \mu_{2n+2}}{(2n+2)!}. \quad (12.6)$$

For real z near zero this is negative, regardless of the parity of n .

12.3 Application to up

For the Rvachev function,

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}(\pi z/2^j).$$

Taking $c = 1/2$ and $Q = 1/4$ in (12.4) gives an exact finite multiscale identity. Its uniform sample-noise amplification is bounded by $K_{\infty}(1/4) \approx 1.96926$, which is notably mild.

More generally, the centered geometric-uniform density u_b in (10.10) has an even transform, and samples at scales $b^k z$ are governed by the interpolation base $Q = b^2$. The cumulants (10.12) determine the coefficients through Bell polynomials, so the filter simultaneously links Gaussian coefficients, Bernoulli numbers, Bell polynomials, and the infinite sinc product.

12.4 Entire Newton reconstruction of all scales

The finite filter recovers the constant term approximately. The infinite Newton series reconstructs the complete scale function.

Theorem 12.2 (Scale reconstruction from a geometric sample ray). *Let Φ be even entire, fix $z \in \mathbb{C}$, and set H_z by (12.2). Then for every $t \in \mathbb{C}$,*

$$\Phi(z\sqrt{t}) = \sum_{j=0}^{\infty} \frac{D_Q^j H_z(1)}{[j]_Q!} \prod_{r=0}^{j-1} (t - Q^r), \quad (12.7)$$

locally uniformly in t . Each coefficient is a finite linear combination of $\Phi(z), \Phi(cz), \dots, \Phi(c^j z)$:

$$\frac{D_Q^j H_z(1)}{[j]_Q!} = \sum_{k=0}^j \frac{(-1)^k Q^{-kj + \binom{k+1}{2}}}{(Q; Q)_k (Q; Q)_{j-k}} \Phi(c^k z). \quad (12.8)$$

At $t = 0$ this gives the exact infinite sum rule

$$1 = \sum_{j=0}^{\infty} \frac{(-1)^j Q^{\binom{j}{2}}}{[j]_Q!} D_Q^j H_z(1). \quad (12.9)$$

Proof. Apply equations (5.4), (5.8) and (5.9) with $A = 1$, base Q , and the entire function H_z . $\square$

This is a strong determinacy statement: at any fixed frequency z , the multiscale values on one geometric ray determine the transform at every complex scale parameter t . For the Rvachev product, this turns the self-similar sample sequence into a complete Newton coordinate system.

13 Algorithms and numerical implementation

The formulas admit several computational realizations. They should not be used interchangeably without regard to conditioning.

13.1 Barycentric interpolation away from zero

For an arbitrary target x , use the second barycentric formula (4.3). The raw weights $b_{n,k}^{A,q} = 1/d_{n,k}^{A,q}$ can span a large range, so in floating-point arithmetic they should be rescaled by a common factor. Only their ratios enter the formula. For large n , compute logarithms of absolute values from (4.1) and attach the sign $(-1)^k$.

At a node, return the corresponding datum directly rather than evaluating a removable 0/0 expression.

13.2 Newton–Horner evaluation

Once the coefficients $c_k = D_q^k f(A)/[k]_q!$ are known, evaluate (5.3) in nested form:

$$\mathcal{I}_n f(x) = c_0 + (x - A)(c_1 + (x - Aq)(c_2 + \cdots + (x - Aq^{n-1})c_n) \cdots). \quad (13.1)$$

This uses $O(n)$ arithmetic per target. Coefficients can be generated from samples by a divided-difference table or by (5.4).

13.3 Origin extrapolation by factor recursion

For the special target 0, the recursion (6.3) is preferable to solving for an interpolating polynomial. In pseudocode:

Algorithm 13.1 (Factorized geometric-comb filter). Given $y_k = f(Aq^k)$ for $0 \leq k \leq n$:

1. Set `work[k] = y[k]`.
2. For $j = 1, \dots, n$, replace

$$\text{work}[k] \leftarrow \frac{\text{work}[k+1] - q^j \text{work}[k]}{1 - q^j}, \quad 0 \leq k \leq n - j.$$

3. Return `work[0]`.

The induced ℓ^∞ error factor of the j th stage is at most $(1 + q^j)/(1 - q^j)$; their product is exactly $K_n(q)$. Thus the recursive algorithm exposes the same stability constant as the explicit row.

13.4 Exact dyadic arithmetic

When $q = 1/2$ and all samples are rational, every weight is rational. For the Fabius sequence, (11.11) and the filter can therefore be implemented with arbitrary-precision integer fractions. This is valuable for separating true cancellation from roundoff. Numerators and denominators grow quickly, so the companion code uses exact fractions only for moderate order and switches to high-precision `mpmath` arithmetic for plots.

13.5 Precision budgeting

Suppose the desired absolute output accuracy is ε and each sample is evaluated with absolute accuracy δ . A sufficient condition is

$$\delta \leq \frac{\varepsilon}{K_n(q)}. \quad (13.2)$$

For fixed $q < 1$, the right side remains comparable to ε as $n \rightarrow \infty$. Near $q = 1$, (8.4) shows that the required number of guard digits grows like

$$\frac{\pi^2}{4(-\log q) \log 10}$$

in the infinite-order regime. Relative precision is subtler: if the samples themselves are superflat, as for $F(2^{-m})$, a small absolute error may still be a large relative error. Exact rational or ball arithmetic is preferable in that case.

14 Numerical experiments

The file `geometric_comb_experiments.py` included in the archive reproduces all data and figures in this section. It uses 140 decimal digits for alternating sums and exact rational arithmetic for the first dyadic Fabius checks. The numerical experiments are confirmations of proved identities, not substitutes for proofs.

14.1 Moment cancellation and factor recursion

For $q = 1/2$ and orders $1 \leq n \leq 20$, the script verifies

$$\sum_{k=0}^n \lambda_{n,k}^{(1/2)} 2^{-kd} = 0, \quad 1 \leq d \leq n,$$

and compares the first residual with $(-1)^n 2^{-n(n+1)/2}$. The largest observed canceled moment is at the level of 10^{-140} , consistent with the working precision. The direct weighted sum and the triangular factor recursion agree to the same scale on the nonpolynomial test function e^x .

14.2 Noise amplification

The data file also compares $K_\infty(e^{-t})$ to the asymptotic (8.4). For the plotted near-one values the relative difference is already below the displayed precision, reflecting the exponentially small modular correction.

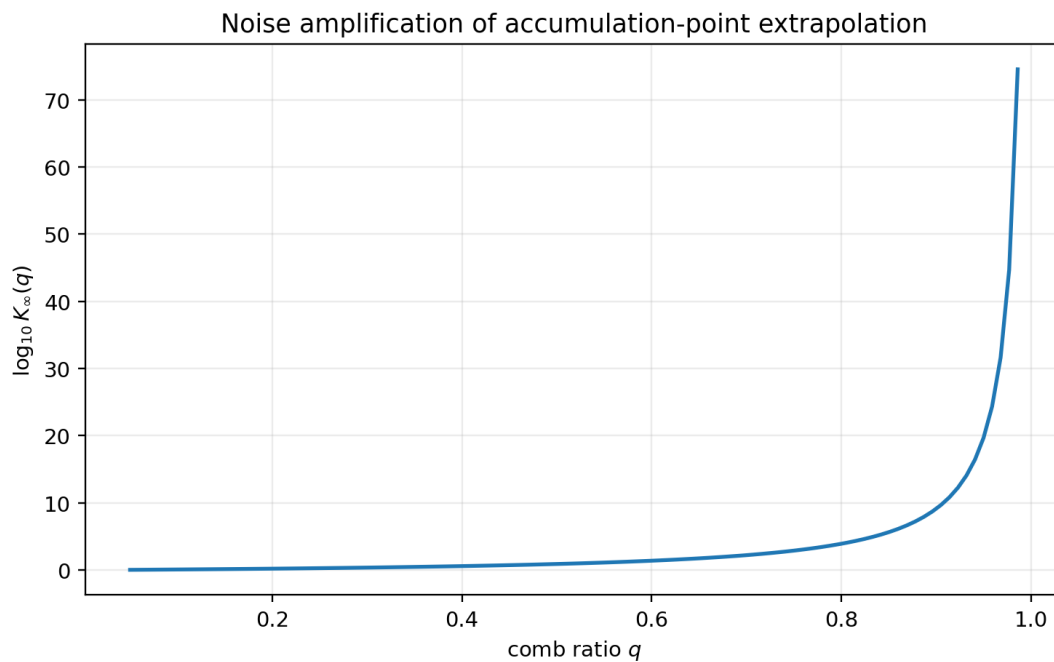


Figure 1: The infinite-order total variation $K_{\infty}(q) = (-q; q)_{\infty}/(q; q)_{\infty}$. The vertical axis is logarithmic in base 10. Stability is excellent for small and moderate q but deteriorates nonperturbatively as $q \uparrow 1$.

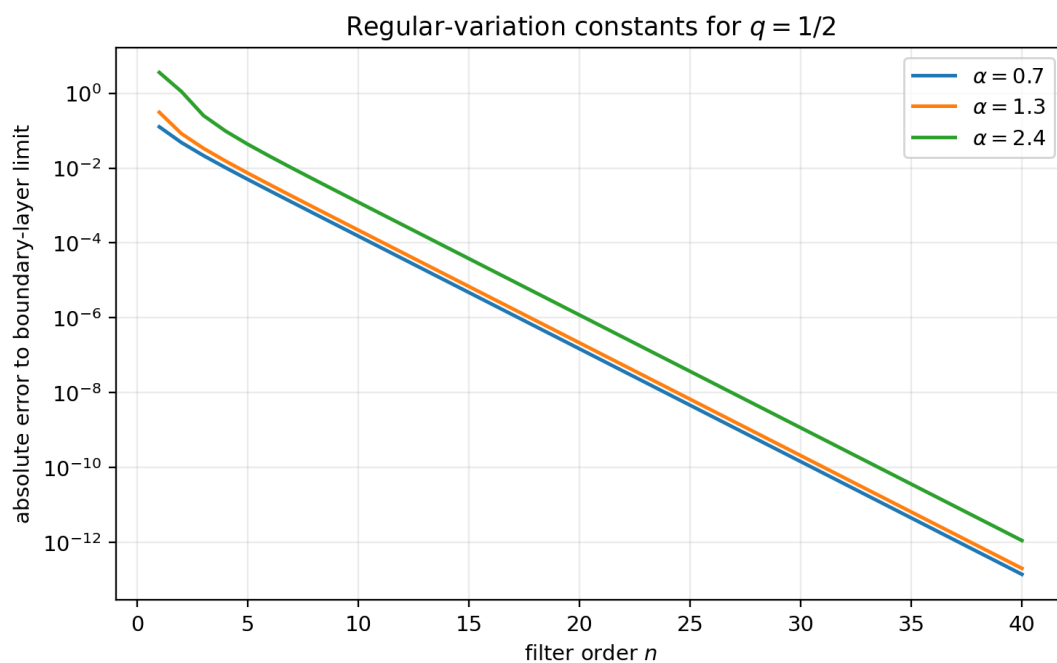


Figure 2: For pure powers x^{α} at $q = 1/2$, the normalized exact multiplier $(q^{1-\alpha}; q)_n/(q; q)_n$ converges geometrically to $C_q(\alpha)$. The three exponents illustrate positive and negative limiting constants.

14.3 Regular-variation constants

For example,

$$\begin{aligned} C_{1/2}(0.7) &\approx 0.2493175488440439431, \\ C_{1/2}(1.3) &\approx -0.1538705960439442305, \\ C_{1/2}(2.4) &\approx 0.2913748265379615867. \end{aligned}$$

At positive integer exponents the limit is exactly zero, in agreement with polynomial cancellation.

14.4 Exact Fabius boundary remainders

For $A = 2^{-(n+1)}$, define

$$R_n := \sum_{k=0}^n \lambda_{n,k}^{(1/2)} F(2^{-(n+1+k)}), \quad B_n := \frac{2^{-n(n+1)}}{(n+1)!}.$$

The theorem gives $(-1)^n R_n > 0$ and $|R_n|/B_n = \mathbb{E}F(S_n) \in (0, 1)$. Exact fraction arithmetic produces the following decimal summary.

Table 2: Normalized exact dyadic Fabius boundary remainders.

n	sign of R_n	$ R_n /B_n$
1	—	0.500000000000
2	+	0.392067901235
3	—	0.285827999336
4	+	0.204528650771
5	—	0.146778506870
6	+	0.106554660592
7	—	0.078481279976
8	+	0.058671832771
9	—	0.044492131277
10	+	0.034188031436

14.5 Reproducibility files

The script writes four CSV files:

File	Contents
<code>moment_checks.csv</code>	canceled moments and recursion comparison
<code>stability_growth.csv</code>	K_∞ , modular approximation, ratio
<code>regular_variation.csv</code>	finite multipliers and limiting constants
<code>fabius_boundary.csv</code>	exact rational filters, bounds, sign checks

The archive includes these files and the three PNG figures, so the LaTeX source can be recompiled without rerunning Python.

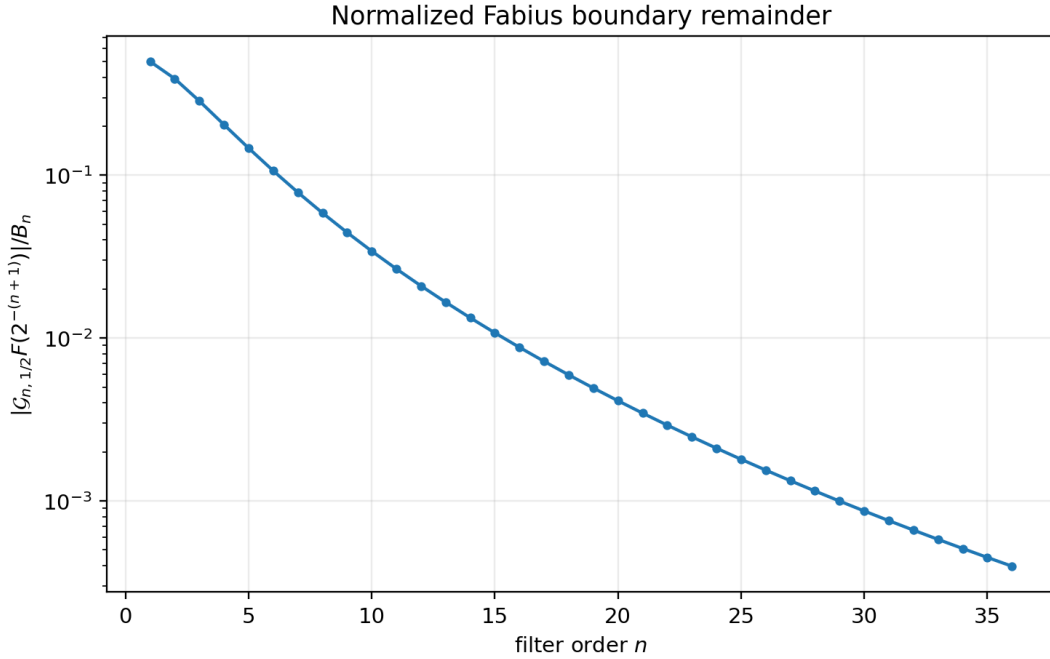


Figure 3: The normalized Fabius boundary remainder $|R_n|/B_n = \mathbb{E}F(S_n)$ through order 36. The decrease is much slower than the already extracted factorial-supergeometric factor B_n ; its finer asymptotics are posed as a conjecture in [section 15](#).

15 Research frontiers and conjectures

15.1 A local limit theorem for geometric B-splines

[equation \(9.8\)](#) proves weak convergence. The explicit spline and limit densities suggest a substantially stronger statement.

Conjecture 15.1 (Local density limit). *For every fixed $0 < q < 1$ and compact interval $K \subset (0, \infty)$,*

$$\sup_{y \in K} \left| \frac{A}{n+2} B_{n,A,q} \left(\frac{Ay}{n+2} \right) - p_q(y) \right| \rightarrow 0, \quad (15.1)$$

where p_q is given by [\(9.11\)](#). The convergence should extend to all derivatives on compact subsets of $(0, \infty)$ after the natural scaling.

A proof might combine the exponential/Dirichlet representation with Fourier or Laplace inversion and uniform control of the denominator $(E_* + \dots + E_n)/(n+2)$. Such a theorem would turn finite interpolation kernels into quantitative approximations of a canonical q -exponential density.

15.2 Fabius–Dirichlet small-value asymptotics

Let

$$\rho_n := \mathbb{E}F(S_n) = \frac{|R_n|}{B_n}.$$

The scaling limit $(n+2)S_n \Rightarrow Z_{1/2}$ suggests that S_n is of order $1/n$, while the Fabius function has a logarithmic-square flatness scale at zero. Random fluctuations of $Z_{1/2}$ should affect lower-order terms but not the leading quadratic logarithm.

Conjecture 15.2 (Leading decay of the normalized Fabius remainder).

$$\log \rho_n \sim -\frac{(\log n)^2}{2 \log 2} \quad (n \rightarrow \infty). \quad (15.2)$$

More precisely, the next terms should be obtainable by combining the sharp Lambert- W endpoint expansion of F developed elsewhere in the ProveIt project with the density (9.11) and a Laplace saddle point.

The experiment in figure 3 is consistent with a slowly converging logarithmic-square law, but orders below 40 are far too small to distinguish the lower-order $\log n \log \log n$ and $\log n$ corrections.

15.3 Second-order regular variation

equation (8.10) gives only the first boundary-layer limit. If $f(x) = x^\alpha L(x)$ and L belongs to a de Haan second-order class, one should be able to expand

$$\frac{(\mathcal{G}_{n,q}f)(A)}{f(Aq^n)} = C_q(\alpha) + A_1(Aq^n)C_{q,1}(\alpha, \rho) + o(A_1(Aq^n)),$$

where A_1 is the auxiliary function and the correction constant is a q -Mellin moment of the signed boundary layer $\beta_r^{(q)}$. This would provide rigorous error estimators for nonanalytic asymptotic extrapolation.

Problem 15.3 (Second-order transfer theorem). *Formulate minimal second-order regular-variation hypotheses and compute the correction constants explicitly from $(qz; q)_\infty / (q; q)_\infty$. Treat the resonant integer cases, where $C_q(\alpha) = 0$, separately; there the correction may become the leading term.*

15.4 The double-scaling transition $q \uparrow 1$, $n \rightarrow \infty$

The fixed-order behavior (8.3) and infinite-order behavior (8.4) do not commute. A natural transition parameter is $n(1 - q)$ or nt when $q = e^{-t}$.

Problem 15.4 (Universal transition kernel). *Determine asymptotics of the row polynomial, its total variation, and the Peano-kernel density when*

$$n \rightarrow \infty, \quad t \downarrow 0, \quad nt \rightarrow \tau \in (0, \infty).$$

One expects a continuum limit connecting geometric-comb extrapolation to confluent Hermite/Taylor extrapolation at A . The relevant asymptotics should involve dilogarithms and a saddle point for finite q -products.

15.5 Optimal filters under analytic priors

The Gaussian filter is algebraically unique when exactly n integer modes are canceled with $n + 1$ samples. With more samples than constraints, stability and bias can be traded.

Problem 15.5 (Minimax geometric extrapolation). *Given $M > n$, choose weights on $A, Aq, \dots, Aq^M$ that preserve constants and annihilate $x, \dots, x^n$, while minimizing either total variation or the worst error over an analytic coefficient-majorant class. Compare the optimizer with truncated, damped, and least-squares modifications of $W_{n,q}$.*

The problem is finite-dimensional and convex for total variation, but its large- M, n structure is likely governed by weighted extremal polynomials on the spectral set $\{q^m : m \geq 0\}$.

15.6 Inverse Fabius and nonlinear combs

The inverse Fabius function has endpoint behavior governed by asymptotic inversion and Lambert W , not a regular Taylor series. A promising strategy is to choose a nonlinear sample comb x_k so that the dominant inverse asymptotic modes become geometric in a transformed coordinate. Confluent modal filters from [section 7](#) can then remove the leading Lambert- W transseries terms.

Problem 15.6 (Lambert-adapted Newton coordinates). *Find a coordinate $u = u(x)$ in which the inverse Fabius germ admits an expansion in modes $e^{-\alpha u} u^s$, construct the corresponding geometric Newton series, and determine whether its interpolation coefficients have a positive kernel or a controlled boundary layer analogous to (8.5).*

15.7 Multivariate geometric combs

Tensor products immediately give interpolation on $\{A_1 q_1^{k_1}\} \times \cdots \times \{A_d q_d^{k_d}\}$, with products of Gaussian rows. More interesting are diagonal or simplex combs with a shared scale index, which interact with multivariate self-similar distributions and box splines. The Dirichlet-average interpretation suggests that multivariate Peano kernels may again have positive probabilistic representations.

15.8 Formalization roadmap

The most suitable theorem sequence for Lean or another proof assistant is:

1. finite q -product identities and the geometric denominator (4.1);
2. row polynomial factorization and moment residuals;
3. Jackson divided differences on monomials, extended by polynomial linearity;
4. operator recursion and total variation;
5. divided-difference remainder plus a library Hermite–Genocchi theorem;
6. Dirichlet moment calculation and the scaling limit via exponential variables;
7. Fabius endpoint derivative induction and boundary sign identity.

The analytic infinite Newton theorem requires more infrastructure, but the finite algebraic and probabilistic core is comparatively modular.

16 Conclusions

Interpolation on a geometric comb is best understood through four mutually reinforcing representations.

First, it is ordinary Lagrange/Newton interpolation with unusually explicit products. Second, it is Jackson calculus: divided differences are normalized q -derivatives, and basis changes are Gaussian-binomial transforms. Third, it is a spectral filter for dilation, making power, fractional-power, and logarithmic asymptotic modes transparent. Fourth, its smooth remainder is a positive B-spline/Dirichlet average whose moments and scaling limit are q -products.

These representations clarify the Fabius/Rvachev connection. Geometrically weighted uniforms produce finite box splines and, at the dyadic value, Thue–Morse coefficients. The Fabius endpoint differential equation converts the Peano remainder into an exact positive expectation with alternating

outer sign. The Rvachev Fourier transform is even, so physical scaling by c becomes geometric interpolation with base c^2 ; at $c = 1/2$ this improves the stability constant from about 8.26 to about 1.97.

The theory also separates two facts that can otherwise be confused. The Vandermonde coefficient problem is severely ill conditioned, but the accumulation-point functional is uniformly stable for fixed $q < 1$. The reason is a universal signed boundary layer concentrated at the smallest samples. That boundary layer controls continuous convergence and regularly varying asymptotics, while the positive Peano kernel controls smooth signs and factorial error bounds.

The resulting framework is broad enough to support modal extrapolation, Fabius endpoint analysis, Fourier moment filters, and future work on inverse functions and double-scaling limits, yet concrete enough for exact rational computation and formal proof.

A Additional derivations

A.1 Partial fractions for the q -exponential perpetuity

For completeness, consider

$$L_q(s) = \prod_{j=0}^{\infty} (1 + sq^j)^{-1}.$$

At the simple pole $s = -q^{-k}$, the coefficient of $(1 + sq^k)^{-1}$ is

$$\begin{aligned} c_k &= \left[\prod_{j \neq k} (1 - q^{j-k}) \right]^{-1} \\ &= \frac{(-1)^k q^{k(k+1)/2}}{(q; q)_k (q; q)_{\infty}}. \end{aligned}$$

Indeed, the factors with $j < k$ give $(-1)^k q^{-k(k+1)/2} (q; q)_k$, and the factors with $j > k$ give $(q; q)_{\infty}$. Since

$$\mathcal{L}^{-1} \left[\frac{1}{1 + sq^k} \right] (y) = q^{-k} e^{-y/q^k},$$

we recover (9.11).

A.2 Equivalence of the power formulas

For an integer $d = n + 1 + r$, (6.5) and (4.7) agree. Indeed,

$$\begin{aligned} q^{dn} \frac{(q^{1-d}; q)_n}{(q; q)_n} &= q^{(n+1+r)n} \frac{\prod_{j=0}^{n-1} (1 - q^{-n-r+j})}{(q; q)_n} \\ &= (-1)^n q^{\binom{n+1}{2}} \frac{(q; q)_{n+r}}{(q; q)_n (q; q)_r} \\ &= (-1)^n q^{\binom{n+1}{2}} \left[\begin{matrix} n+r \\ n \end{matrix} \right]_q. \end{aligned}$$

This calculation is often useful when moving between Mellin exponents and integer Taylor moments.

A.3 A generating function for the smooth remainder on exponentials

Take $f(x) = e^{tx}$ in (9.2). Using (9.6),

$$\begin{aligned} (\mathcal{G}_{n,q}e^{t\cdot})(A) - 1 &= \frac{(-1)^n A^{n+1} q^{\binom{n+1}{2}} t^{n+1}}{(n+1)!} \mathbb{E} e^{tY_{n,A,q}} \\ &= (-1)^n A^{n+1} q^{\binom{n+1}{2}} t^{n+1} \sum_{r=0}^{\infty} \frac{A^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q}{(n+r+1)!} t^r. \end{aligned} \quad (\text{A.1})$$

The same series follows from (4.9). The equality is a useful consistency check between the analytic-tail and B-spline pictures.

B Companion-code notes

The Python source intentionally avoids black-box interpolation routines. Every quantity is implemented from the formulas proved in the report. The key functions are:

origin_lagrange_weights
implements (4.4);

recursive_filter implements (6.3);

stability_factor_infinite
computes (8.2);

regular_variation_constant
computes (8.10);

fabius_dyadic_values_exact
implements (11.11) with rational arithmetic;

fabius_boundary_filter_exact
checks (11.8) exactly.

The script can be rerun from the archive directory with

```
python geometric_comb_experiments.py
```

It overwrites only the listed CSV and PNG outputs.

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